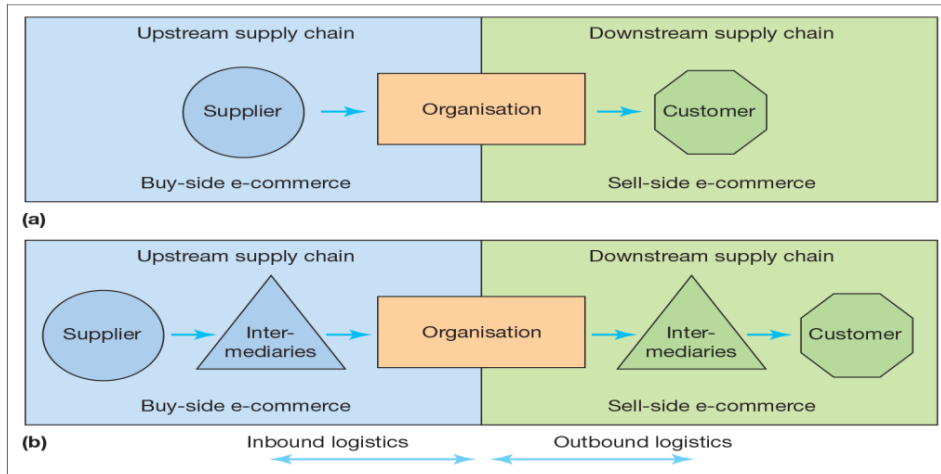


Ch2_Supply Chain Risk Management

一、Review of SCM



A **supplier** is a person or a company that supplies large amounts of **raw materials** to manufacturing businesses.

Supply chain management (SCM) The coordination of all **supply activities** of an organisation from its suppliers and partners to its customers.

Upstream supply chain Transactions between an organisation and its suppliers and intermediaries, equivalent to buy-side e-commerce.

Downstream supply chain Transactions between an organisation and its customers and intermediaries, equivalent to sell-side e-commerce.

Upstream Supply Chain 上游供应链

- 企业通过电商平台采购原材料、零部件等
- Inbound Logistics (入境物流)：从外部进入企业的物流

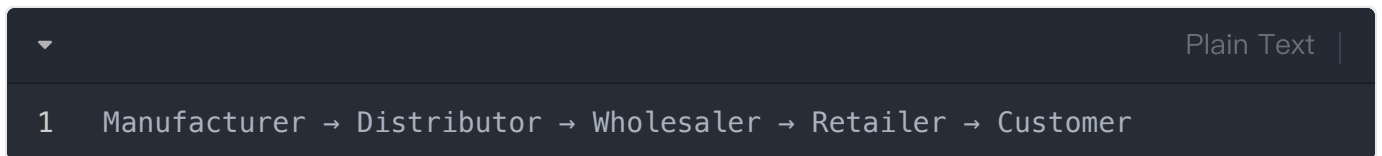
Downstream Supply Chain 下游供应链

- 电商平台销售产品给终端客户
- Outbound Logistics (出境物流)：从企业发出到客户的物流

Intermediaries – distribute, promote, store, or sell goods.

Manufacturer	Produces the goods from raw materials
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Distributor	Handles logistics, regional coverage , and inventory for multiple wholesalers
Wholesaler	Buys in bulk from distributors, sells in smaller batches to retailers
Retailer	Sells directly to consumers
Customer	Final recipient and user of the product



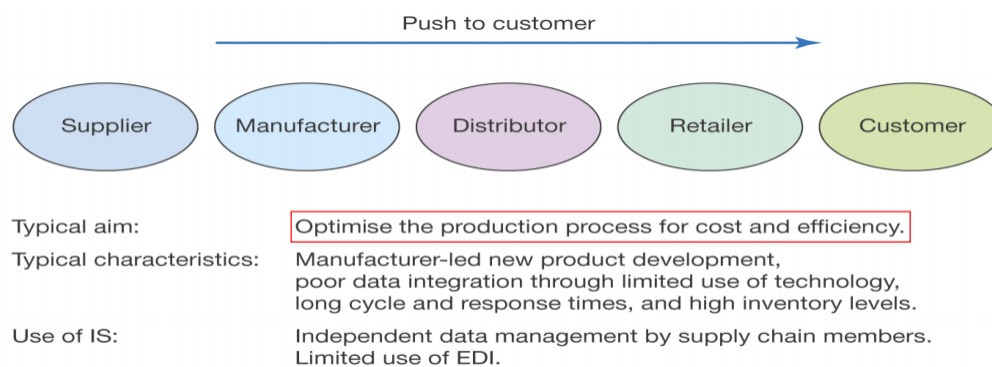
Each arrow represents a **transaction between distinct organizations**, and each node is an **organizational role** in the supply chain:

So when you ask “where is the organization?”— the answer is: **everywhere**. The supply chain is a network of organizations, each performing a specialized function.

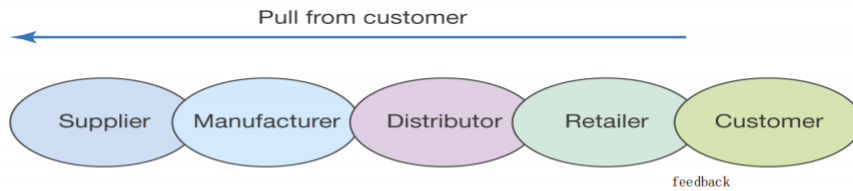
A **distributor** acts as an **exclusive or authorized intermediary for a manufacturer**, promoting and selling products to downstream businesses like wholesalers or retailers, often **taking ownership** of the goods. They handle regional or national delivery -- 区域物流、库存管理和市场覆盖

A **wholesaler** buys in bulk **from distributors** or **directly from manufacturers** and sells to **retailers**, who then sell to consumers.

A **retailer** buys products from wholesalers or distributors and sells them directly to the end **consumers**, typically in smaller quantities and at a higher markup.



(a)



Typical aim:

Enhance product and service quality.

Typical characteristics:

Market research driven, technology used to achieve research and data integration, short cycle and response times, low inventory levels.

Use of IS:

Integrated internal systems, information sharing between supply chain members. Extensive use of EDI and e-commerce, often through B2B exchanges and intermediaries.

(b)

Vertical integration	Vertical disintegration (disaggregation)	Virtual integration
Characteristics: - Majority of manufacture in-house - Distant relationships with suppliers	Characteristics: - Move to outsourcing - Network of suppliers	Characteristics: - Total reliance on linked third parties - Close relationships with suppliers
Applications: Specialised or proprietary production	Application: Cost reduction and focus on core capabilities	Application: Rapid market penetration (dot-com approach)

自产自销

集成别人的功能 inside the platform

1. Vertical integration 垂直整合

- in house
- Relationship with suppliers – Distant
- 专有或定制化生产 Specialised or proprietary production
- High control, low flexibility
 - flexibility as a response to uncertainty and change.

2. Vertical disintegration 垂直解构（分散化）

- Move to **outsourcing**（外包生产）
- Suppliers network
- 降低成本，专注核心功能 – R&D high cost ↑
- Control ↓, efficiency ↑

3. Virtual integration 虚拟整合

- 完全依赖外部合作伙伴
- 与供应商建立紧密关系
- 快速进入市场

- High flexibility, High dependency

Strategy 类型	Risk Response 应对风险	Example 举例
Vertical Integration 垂直整合	❌ Slow to adapt 应变慢，风险集中	Traditional auto manufacturers 传统汽车制造商
Vertical Disintegration 垂直解构	⚠️ Can shift suppliers 可更换供应商，降低部分风险	Apparel brands outsourcing to Asia 服装品牌外包至亚洲
Virtual Integration 虚拟整合	✅ Rapid adaptation 快速应变，风险分散	E-commerce platforms with API-linked suppliers 通过API连接供应商的电商平台

▼ 通过API连接供应商的电商平台

假设你在京东开设一家店铺，销售某品牌的耳机：

- 如果你通过API连接了该品牌的供应商系统：
 - 当供应商库存变动，你的店铺自动更新库存信息。
 - 当客户下单，订单信息自动发送给供应商进行发货。
 - 如果产品价格调整，你的页面也同步更新。

垂直整合：控制力强，但一旦市场或技术发生变化，调整成本高，风险集中在内部。

垂直解构：通过外包获得一定灵活性，但仍依赖稳定的供应商网络。

虚拟整合：通过数字化连接多个合作方，快速响应市场变化，但也面临协调和信任风险。

▼ 为什么 virtual integration 更加灵活?

1. 实时数据流动

- **传统方式**：信息传递依赖纸质文件、电话或面对面沟通，滞后严重。
- **数字化方式**：通过API、ERP系统、云平台等，供应链各节点可以**实时共享库存、订单、运输状态等数据**。
- **✅ 优势**：快速响应市场变化，及时调整生产或采购计划。

2. 动态合作网络

- **传统方式**：合作关系固定，切换供应商或渠道成本高。
- **数字化方式**：平台化结构允许企业**快速接入或替换合作方**，如电商平台上的第三方卖家或物流服务商。
- **✅ 优势**：应对突发事件（如疫情、断供）时更具弹性。

3. 模块化架构与自动化决策

- 数字化系统支持**模块化配置**，比如自动补货、**智能定价**、预测性维护等。
- **✅ 优势**：企业可以根据不同市场或客户需求，快速调整产品组合或服务流程。

4. 可视化与预测能力增强

- 借助数据挖掘与AI分析，企业可以提前预测风险（如供应中断、需求波动）。
- **✅ 优势**：从“被动应对”转向“主动规划”

二、SCM RISK

What motivated RISK management in supply chain?

- **Global competition** – Supply chains stretch across borders – involving multiple vendors, regulations and geopolitical risks.
 - A disruption in one country can ripple across the entire chain.
- **Technological change**
 - Cybersecurity
 - The usage of advanced technology leads to difference
- **Continual search for competitive advantage**
 - Businesses constantly seek ways to differentiate—faster delivery, lower costs, better quality.

- Pushing for leaner operations (浪费↓) or just-in-time inventory can increase vulnerability to disruptions.
 - JIT – 保持最低库存, 降低成本。

4 risk reduction strategies used within supply chains

1. SCM risk **identification** – different types of risk

Operation risks – Uncertainties for **supply chain elements** (customer demand – 需求变化 , supply – 供应商交付延迟 and cost – 成本变化 如, 税务影响)

Disruptions risks 中断风险 – 突发事件

- Natural disaster – flood
- economic crises – shifting market price , currency reevaluation

2. Estimation of **likelihood** of each event

估算每种事件发生的**可能性** ---> 需要结合历史数据、预测模型、行业经验

“**event**” refers to any **specific incident or occurrence** that could potentially disrupt the normal flow of the supply chain.

3. Assessment of **potential loss** from major disruption – **Quantitative**

- Political events
- Product availability
- Distance from source
- Industry capacity
- Demand fluctuation
- Change in technology
- Change in labor markets
- Financial instability
- Management turnover

Category	Example	Impact
Political	Political events	Trade restrictions, sanctions, instability
Operational	Product availability, distance from source	Delays, cost increases
Market	Industry capacity, demand fluctuation	Supply–demand mismatch

Technological	Change in technology	Obsolescence, adaptation cost
Labor	Change in labor markets	Talent shortages, strikes
Financial	Financial instability	Credit risk, currency volatility
Organizational	Management turnover	Strategic drift, decision paralysis

4. Identification of strategies to reduce risk

风险规避旨在彻底消除风险发生的可能性，并消除其对组织造成不利影响的能力。

风险缓解并非消除风险发生的可能性，而是将其影响降低到可接受的水平。

- **Risk Avoidance – 风险规避**

- 前置性策略，消除风险发生的可能性

- Insurance – finance risks

- Customer – Loss of customers due to supply chain disruption

- 通过提高交付可靠性、透明沟通来降低客户流失风险

Supply Chain Alliances	与供应商建立长期合作关系
E-procurement	电子采购系统，提高效率与透明度
Just-in-Time Delivery	精准交付，减少库存成本
Increased Coordination	加强协调与信息共享

- E-procurement is a **buyside** function, as it focuses on a company or organization purchasing goods and services through online platforms and digital systems, rather than selling them

- 简化采购流程、降低成本和提高组织效率

- JIT – 最大限度地降低库存仓储成本并提高整体效率的物流策略

- JIT的核心原则围绕着“拉动”而非“推动”生产的概念。换句话说，商品是根据实际需求生产或交付的，而不是根据预测需求推入市场。

- **最大限度地减少浪费：** JIT 旨在消除供应链中的浪费。这包括库存过剩、生产过剩、等待时间和不必要的运输。通过“准时制”运营，企业可以减少浪费行为并优化资源。

- 旨在通过消除任何不会给最终产品或服务增加价值的东西来简化流程、降低成本并提高效率。

- “produce only when needed,” often triggered by customer demand

- JIT works best when the supply chain is fast, reliable, and responsive.



JIT Simplified: “Customer Need → Then Produce”

Yes, that’s the core logic:

- No stockpiling
- No early production
- Only act when there's a confirmed need

But here's the catch:

If production or delivery takes too long, JIT can backfire.

Dimension	Impact	Explanation
Vulnerability	▲ Increases	With minimal buffer stock, any delay (e.g., supplier issue, transport disruption) can halt operations immediately
Likelihood of Risk	▲ May increase	JIT relies on perfect coordination; even small fluctuations in demand, supply, or logistics can trigger disruptions
Severity of Impact	▲ Often higher	Because there's no cushion, even minor issues can cause major operational breakdowns

- **Risk mitigation 如何缓解风险?**

- 风险缓解发生在实际风险事件发生之前因为它涉及采取主动措施来降低潜在威胁的可能性或影响

1. 管理**不确定性 uncertainty** – 供给 supply、需求 demand、产品 product management、信息 information management.
2. **Network design** can enable better control of supply risks
 - 供应链的结构设计（比如是否有多个供应商、是否有区域分布）会影响风险控制能力
3. Strategies such as **product pricing** can control demand to a degree
 - 通过定价策略控制需求 – 通过调整价格，可以在供应紧张时抑制需求，或在库存过剩时刺激购买。
4. System can provide greater information visibility across supply chain members – 信息共享是降低风险的关键。越多成员能实时看到数据，越能快速响应。

5 interrelated issues in supply management

供应管理中5个关键、相互关联的问题：

1. Supply network design – 供应网络设计

2. Supplier relationships

3. Supplier selection process – 如何评估和筛选合适的供应商，包括成本、质量、交货期、信誉等因素。

➤ Include the following criteria

- Quality personnel
- Quality procedure
- Concern for quality
- Compare history
- Price relative to quality
- Actual price
- Technical performance
- Delivery history
- Technical assistance
- Production capability
- Manufacturing equipment

- a. Quality personnel (人员素质) – 供应商的员工是否具备专业技能、经验、质量意识
- b. Quality procedure – 供应商是否有系统化的质量管理流程
- c. Price relative to Quality (性价比)
- d. Delivery History (交货记录) – 是否按时交货
- e. Technical Assistance (技术支持) – 供应商是否提供安装、维护、培训等技术服务

4. Supplier order allocation – 如何将订单分配给不同供应商

供应商订单分配”中的**操作风险** (Operational Risks)包括：

Demands (用户需求波动)、**Supply yields** (供应产出不确定性 – 原材料、设备、人力...)、**Lead times** (交货周期不确定性)、**Cost** (成本 – 原材料价格、运输费用、汇率...)

5. Supply contracts – 合同类型 (固定价格、可变价格、期权合同等) 如何影响供应链的灵活性和风险。

E-commerce security

1. Pros: 电子采购帮助供应链联盟伙伴 supply chain alliance partners 显著降低成本 **cost reduction**
2. Cons: Hackers or competitors gaining information over the Web.
3. Solution: Better hiring and training of employees; Certification and auditing of outsourcing partners.

SCM Risk Strategy

1. Flexible Supply Base

Mitigates risk from sole sourcing through **multiple suppliers**.

通过多个供应商来降低“单一来源”的风险。

2. Make and Buy Strategy

Including **external** production as an **alternative source**.

结合自制与外购，确保在内部产能不足时有外部支持。

3. Economic Supply Incentives

Used even if production cannot be shifted.

即使无法转移生产，也可通过经济激励保障供应

4. Flexible Transportation

Provides assurance of delivery.

使用多种运输方式或备用物流渠道，确保交付可靠性。

5. Strategic Stock

Safety stock without excessive inventory cost.

设置关键物料的战略库存，避免全面囤货带来的高成本。

6. Dynamic Pricing and Promotions

Forms of revenue management.

通过动态定价和促销策略来调节需求，缓解供应压力。

7. Postponement

Improves the ability to manages supplies and accomplishes greater product flexibility;

Delays product differentiation until closer to the time of sale.

延迟产品定制或差异化，直到更接近销售时点，以应对需求变化。

像服装品牌先生产白色T恤，等订单确定后再染色。

8. Dynamic Assortment Planning 动态商品组合规划

based on influencing consumer product demand by display location;

通过调整商品在货架或页面上的展示位置，影响消费者的购买偏好和需求分布。

电商平台首页推荐、实体店货架布局优化。

9. Silent Product Rollover 静默产品替换

involves slow leading of new products without formal announcement.

不进行正式宣传的情况下，逐步引入新产品，减少库存冲击和市场不确定性。

Discussion Questions



- What are the difference between vertical integration, vertical disintegration and virtual integration ?
- Which factors motivated risk management in supply chains ?
Provide examples for each of the factors.
- What are the key four risk management in supply chain?
Briefly explain each of them.
- Which criteria should be included in the supplier selection process?
- Why e-commerce security is so important in supply chain?
How to reduce the risk for the e-commerce security?
- Analysis how to apply 9 robust supply chain strategies into JD company in mainland China.